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DEVER RTD EAGLE P3

Communication System

Software Verification & Validation Plan (SVVP)

DISTRIBUTION	
TOTAL	

REVISION							
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COMMUNICATION SYSTEM FOR DENVER RTD EAGLE P3 PROJECT

WOOJIN Industrial Systems Co., Ltd.

REVISION RECORD

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1. Purpose

This software(called "SW" here after) verification and validation plan provides a specific method and procedure for the verification process to ensure that work products or products or product components from the engineering activities from the start to completion of project are in compliance with requirements, and validation process to ensure that needed function performs properly in the intended environment.

This Verification and Validation Plan applies to software requirements, software design, and software implementation phases of software development life cycle.

The audience for this document consists of the following groups and individuals:

- WOOJIN's software engineering group, project management and software quality assurance
- Rotem and RTD representatives

2. Reference Documents

This document shall be used in conjunction with the following documents and publications. If the reference documents or publications are superseded by an approved revision, the most recent revision shall apply.

- IEEE STD 1012 for SW Verification and Validation
- IEEE 829 Standard for Software Test Documentation

3. Terms, Definition and Acronym

3.1 Terms and Definition

The definition of terms is described as follows:

Terms	Definition
Verification	Activity to ensure that work products, product or product components comply with requirements and customer is not involved generally. Verification is an incremental process covering the whole project life cycle starting from the requirement verification, verifying the intermediate work products at the end of each phase and completing with the verification of the product completed. "You built it right"
Validation	Activity to demonstrate that work products, products or product components are functioning in the intended environment and normally customer is involved. Validation activity uses a similar method used in verification and is sometimes done at the same time with verification using a part of same environment. "You built the right thing"
Peer	Generally refer to the person who is not the manager of the work product author. When necessary, the manager of the author may become the peer of the work product author for the review of the work product.
Peer Review	Review activity performed by the peer of the work product author to detect defects of the work product through the whole phase of the product lifecycle. Cover informal review for which customer is not involved, and Formal Review for which customer jointly participates.
V&V personnel	Person who has responsibility and authority to establish V&V plan and to perform V&V activity.
Inspection	A kind of peer review led by the peer review leader where review meeting is held according to the pre-defined role to identify defects or work products which has been distributed in advance. Benefit: maximize the identification of defects Weakness: Separate amount of efforts for review planning and pre-review.
Walkthrough	A kind of peer review led by the author of the work product and review meeting is held without pre-review of work products. Benefit: minimize the time for review. Weakness: work product author may be biased.
Desk Check	A kind of peer review done with the least effort and this method is used for relatively non-critical products or environmental restriction is inevitable; Desk check may not be recorded.

Terms	Definition
Joint Review	A formal meeting to have approval or to invite opinion of the final user, customer and stakeholders, and formally check that H/W, S/W from the phase is equipped with the required quality.
Defect	Incompleteness or various abnormal performance related items of work products or product which prevents system or component perform the required functions.
Engineering Manager	The manager of the Engineering department at Woojin.
Project Manager	The manager of the project as designated by management at Woojin.
QA Group	One or more representatives of the Quality department at Woojin as designated by the Quality Manager.
Software Configuration Item	An aggregation of software that is designated for configuration management and treated as a single entity in the configuration management process. The Software Configuration Item is distributed, installed, and version tracked as one item.

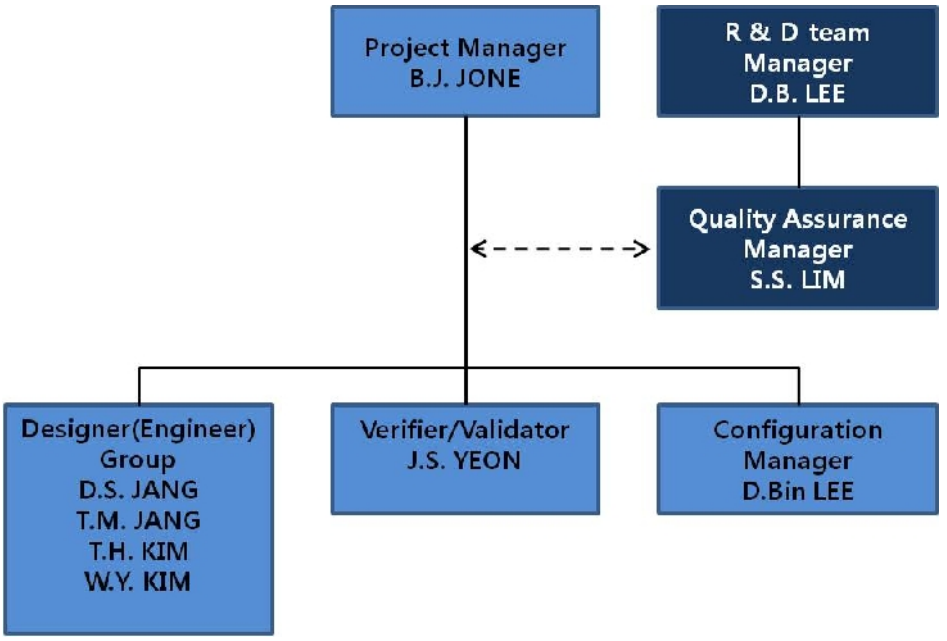
3.2 Acronym

Acronym used in the SW VV plan is as follows:

Acronym	contents
SW	Software
COTS	Commercial-Off-The-Shelf
SDD	Software Design Description
SRS	Software Requirements Specification
SVVP	Software Verification and Validation Plan
SVVR	Software Verification and Validation Report
V&V	Verification and Validation

4. V & V Summary

4.1 Organization



4.2 Roles and Responsibility

Role	Responsibility	Work products
Customer	● Review V&V plan ● Check V&V plan Reported on the V&V activity results and corrective action results	Approval
Project Manager	● Establish and document V&V plan. ● Confirm the V&V activity result and report to the customer. ● Confirm the quality of V&V activities.	SVVP and SVVR, Approval
QA Manager	● Review V&V plan ● Approves V&V plan. ● Review V&V results independently from project view ● Participate in the training support and activity for the performing V&V process as needed	Approval
Verifier/Validator	● Verification and Validation activities of the project work products ● Author and maintain test plan and test report ● Control test records and track defects	SVVR
Software Engineering Group	● Develop the configuration items by staged and request the changes of configuration items if necessary. ● Establish the changes for the configuration item. ● Perform the corrective action of nonconformance caused configuration audit results. ● Participate in the V&V activities. ● Provide work products needed for V&V activities.	SVVR

4.3 Schedule

The project software life cycle follows Woojin's Engineering Process. Verification milestones correspond with the project milestones at the end of each development phase.

Phase	V&V Activity	Result	Schedule/ Participant
Planning	SW Planning V&V activity	Software Planning Verification and Validation Report	2011. 6. 15 / B.J. JONE, J.S. YEON
Requirement Analysis	SW Requirement V&V activity	Software Requirements Verification and validation Report (Updated)	2011. 7. 31 / B.J. JONE, J.S. YEON
Design	SW Design V&V activity	Software Design Verification and validation Report (Updated)	2011. 8. 31 / B.J. JONE, J.S. YEON
Implementation	SW Implementation V&V activity	Software Implementation Verification and Validation Report (Updated)	2011. 12. 31 / B.J. JONE, J.S. YEON
Test	SW Test V&V activity	Software Test Verification and Validation Report (Updated)	2012. 2. 28 / B.J. JONE, J.S. YEON
Maintenance Release	N/A	N/A	N/A

Refer to section 6.0 for more detail information.

4.4 Master Schedule

Phase	Beginning	Complete	2011								2012			
			5	6	7	8	9	10	11	12	1	2	3	4
Plan	11/05/01	11/06/15												
Requirements	11/06/16	11/07/31												
Design	11/08/01	11/08/31												
Implementation	11/09/01	11/12/31												
Test	12/01/01	12/02/28												
Maintenance	12/03/01													

➢ Major Milestone

- 2011.05.01: Project Beginning
- 2011.07.31: Requirements Complete
- 2011.08.31: Design Complete
- 2011.12.31: Product Complete
- 2012.02.28: Test Complete
- 2012.03.01: Maintenance Beginning

4.5 Resource Summary

Human resources for software verification tasks consist of members of Woojin's Engineering Department, Management and QA Group, and Verifier/Validator
Documentation tool is Microsoft Word 2003/2007.

Software V&V Check List is based on ANS/IEEE 1012 and use for V&V activity.

Type	Contents	Comment
Efforts	Refer to 4.1 organization, 4.4 Schedule	
Tool used	Process checklist, Work product Check list, Test Tool	

4.6 Tools, Techniques and Methods

Tools and technique used for this project is as follows.

Type	item
Documentation tool	MS Office 2003/2007
Analysis	Case Studio 2
V&V	Checklist

Human efforts shall be used for documents reviews and Software test.

4.6.1 Document Review

Document review verifies accuracy and completeness of the information contained in the document. The objects of documents review shall be Software Requirements Specification and Software Design Specification.

4.6.2 Testing

Testing shall be performed from the viewpoint of software requirements.

5. Software Integrity Level

The safety Integrity Level (SIL) of the PA/PIS system is 0 (Non safety-related) for all PA/PIS hardware and software as described in the DENVER RTD Communication system SSIL (Software Safety Integrity Level) Justification document.

6. V&V Processes

V&V Process supports Planning V&V, Concept V&V, Requirement V&V, Design V&V, Implementation V&V, Test V&V, Installation and Checkout V&V and maintenance V&V
For more details about V&V activities, WOOJIN will follow the table 1 of IEEE-STD-1012.

6.1 Planning

Phase	INPUT	Task	OUTPUT	Deliverable
Planning	• SCMP • SQAP • SVVP • SPMP • SW V&V Check List	Technique/Management activity should be reviewed for software to check, identify, monitor to assure software required quality aim. **Method** ✓ Document Review	• SW V&V Check Result • SVVR	SVVR

6.2 Requirement Analysis

Phase	INPUT	Task	OUTPUT	Deliverable
Requirement Analysis	• SRS • SFD • SW V&V Check List	SRS should be reviewed to meet the requirements of Technical Specifications and Procurement Technical Specification. **Method** ✓ Document Review	• SW V&V Check Result • SVVR revision • SRTM Revision (when required)	• SVVR revision • SRTM Revision (when required)

6.3 Design

Phase	INPUT	Task	OUTPUT	Deliverable
Design	• SDD • SW V&V Check List	Software Design Specification should be reviewed to meet the requirement of structured analysis/design. **Method** ✓ Data Flow Diagram Review ✓ Structure Chart Review ✓ Detailed Design Review	• SW V&V Check Result • SVVR revision • SRTM Revision (when required)	• SVVR revision • SRTM Revision (when required)

6.4 Implementation

Phase	INPUT	Task	OUTPUT	Deliverable
Implementation	• SW Source Code • SW V&V Check List • SW Source Code check List	Software Source Code should be reviewed to be implemented on the basis of Software design. **Method** ✓ Source Code Peer Review	• SW V&V Check Result • SVVR revision • SRTM Revision (when required)	• SVVR revision • SRTM Revision (when required)

6.5 System Integration Test

Phase	INPUT	Task	OUTPUT	Deliverable
System Integration Test	• Test Plan • System Integration /Procedure/ Report • SW V&V Check List	System Integration test procedure/report should be reviewed to verify whether operated correctly that SRS was intended. **Method** ✓ CCP Test	• SW V&V Check Result • SVVR revision • SRTM Revision (when required)	• SVVR revision • SRTM Revision (when required)

6.6 Type Test

Phase	INPUT	Task	OUTPUT	Deliverable
Type Test	• Type Test Procedure/ Report • SW V&V Check List	Process each type of performance test by certain format of the competent authorities, loading software which has completed the system integration test on the hardware. **Method** ✓ Factory Test	• SW V&V Check Result • SVVR revision • SRTM Revision (when required)	• SVVR revision • SRTM Revision (when required)

6.7 Complete Car Test

Phase	INPUT	Task	OUTPUT	Deliverable
Complete Car Test	• Complete Car Test Procedure/ Report • SW V&V Check List	Verify whether the hardware design which has completed the type test operates properly as intended on the rail vehicle. Method ✓ Field Test	• SW V&V Check Result • SVVR revision • SRTM Revision (when required)	• SVVR revision • SRTM Revision (when required)

6.8 Main Line Test

Phase	INPUT	Task	OUTPUT	Deliverable
Main Line Test	• Main Line Test Procedure/Report • SW V&V • Check List	Implement test on the high speed line by temporarily operation command to verify the accurate performance and function of vehicle system. Method ✓ Field Test	• SW V&V Check Result • SVVR revision • SRTM Revision (when required)	• SVVR revision • SRTM Revision (when required)

6.9 Release and Maintenance

Phase	INPUT	Task	OUTPUT	Deliverable
Release and Maintenance	• N/A	Receive the requirements and problems during the software operation from the contractor. Compose a release package including the changed documents and re-release it when the related problem has been solved. Method ✓ N/A	N/A	• N/A

7. V&V Report, and Documentation Requirement

The Verification and Validation Reports contain various software verification and validation task results for the project, plus a summary of the anomalies and a description of how the anomalies were resolved.
The Verification and Validation Report for software requirements specification, software design, Software Source Codes and Software Test shall be created.

8. V&V Administrative Requirements

8.1 Anomaly Resolution and Reporting

Anomalies that affect a software configuration item in a released product shall be recorded in the Software Verification and Validation Report.

The Project Engineer is responsible for ensuring that appropriate actions for resolution are initiated, involving one or more of the following activities:

- Modification of the software design
- Modification of a test procedure
- Modification of the software
- Generation of a suitable report detailing why the anomaly is acceptable

8.2 Control Procedures

All inputs and outputs to verification and validation tasks are subject to configuration management standards as defined in the Software Configuration Management Plan.
All verification and validation tasks are subject to QA review and audit at any time throughout the life of the project.

8.3 Standard, Practices and Conventions

Standards applied to the verification and validation activities, refer to section 2 of this document.